

Bézout Domain and Noetherian Domain

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Recall: Definitions in Ring

Definition

Definition

A Commutative Ring R with 1 is called an **Integral Domain** if:

R has no zero divisor.

That is, for any $a, b \in R$, $ab = 0$ implies either $a = 0$ or $b = 0$.

Definition

In an integral domain R , a subring $I \subset R$ is called an **Ideal** if:

Given $r \in R$ and $a \in I$, ra is contained in I

Proposition

Let R be an integral domain, and $A \subset R$ be a subset. Then,

$$(A) \stackrel{\text{def}}{=} \bigcap_{\substack{A \subset I \\ I \text{ is ideal}}} I = RA = \{r_1 a_1 + \cdots + r_n a_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\}$$

Bézout Domain

Definition (Bézout Domain)

An integral domain is called a **Bézout Domain** if it satisfies:

for any $a, b \in R$, there exists $d \in R$ such that $(a, b) = (d)$

Example

In the polynomial ring $\mathbb{Z}[x]$,

$$(2, x) \neq (d)$$

for all $d \in \mathbb{Z}[x]$. Thus, $\mathbb{Z}[x]$ is **Not** a Bézout domain.

Bézout Domain: Every finitely generated ideal is principal

Theorem

Let R be an integral domain. Then,

R is a Bézout Domain

if and only if

Every finitely generated ideal in R is principal

proof sketch.

Observe that

$$\begin{aligned}(a_1, a_2, \dots, a_n) &= (a_1) + (a_2) + \dots + (a_n) = (d_1) + (a_3) + \dots + (a_n) = \\ &\dots = (d_{n-1})\end{aligned}$$

□

Noetherian Domain

Noetherian Domain

Definition (Noetherian Domain)

An integral domain R is called a **Noetherian Domain** if it satisfies:

For every sequence of ideals $\{I_n\}_{n=1}^{\infty}$ such that

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$$

, there exists $N \in \mathbb{N}$ such that $I_n = I_N$ for all $n \geq N$.

Such a condition is called the **Ascending Chain Condition** on ideals.

Example

In the infinitely many variables polynomial ring $\mathbb{R}[x_1, x_2, \dots]$,

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \dots$$

never stabilizes.

Noetherian Domain: Every ideal is finitely generated

Theorem

Let R be an integral domain. Then,

R is a Noetherian Domain

if and only if

Every ideal in R is finitely generated.

Principal Ideal Domain

Principal Ideal Domain

Definition

An integral domain R is called a **Principal Ideal Domain** if:

Every ideal in R is principal

Theorem

Let R be an integral domain. Then:

R is a **Principal Ideal Domain**

if and only if

R is **Bézout Domain** and **Noetherian Domain**

End
